

Multiple Choice

1. **Answer:** E

Options: A) $L\{t\} = (2 / s^3)$; B) $L\{t\} = (1 / s^3)$; C) $L\{t\} = s^2$; D) $L\{t\} = (1 / (2 s^2))$; E) $L\{t\} = (1 / s^2)$

Note: Correct option: E - $L\{t\} = (1 / s^2)$

2. **Answer:** A

Options: A) ROM; B) FPGA; C) PLA; D) PLD; E) EEPROM

Note: Correct option: A - ROM

3. **Answer:** D

Options: A) 0.785 J; B) 0.211 J; C) 0.125 J; D) 0.623 J; E) 0.956 J

Note: Correct option: D - 0.623 J

4. **Answer:** A

Options: A) 0.3 msec; B) 0.8 msec; C) 0.6 msec; D) 0.25 msec; E) 0.1 msec

Note: Correct option: A - 0.3 msec

5. **Answer:** A

Options: A) 8,762.3 psi; B) 4,321.0 psi; C) 25,678.9 psi; D) 6.25 lbs; E) 30,000 psi

Note: Correct option: A - 8,762.3 psi

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '-0.6008, -2.4486, 0.0247 + j 0.824, 0.0247 - j 0.824'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '500 MVA'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 0.125. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 1.00 J. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key